



DIGITAL VLSI DESIGN

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DIGITAL VLSI DESIGN

PART 1

MOS Inverters (Static Characteristics)

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MOS Inverters (Static Characteristics)

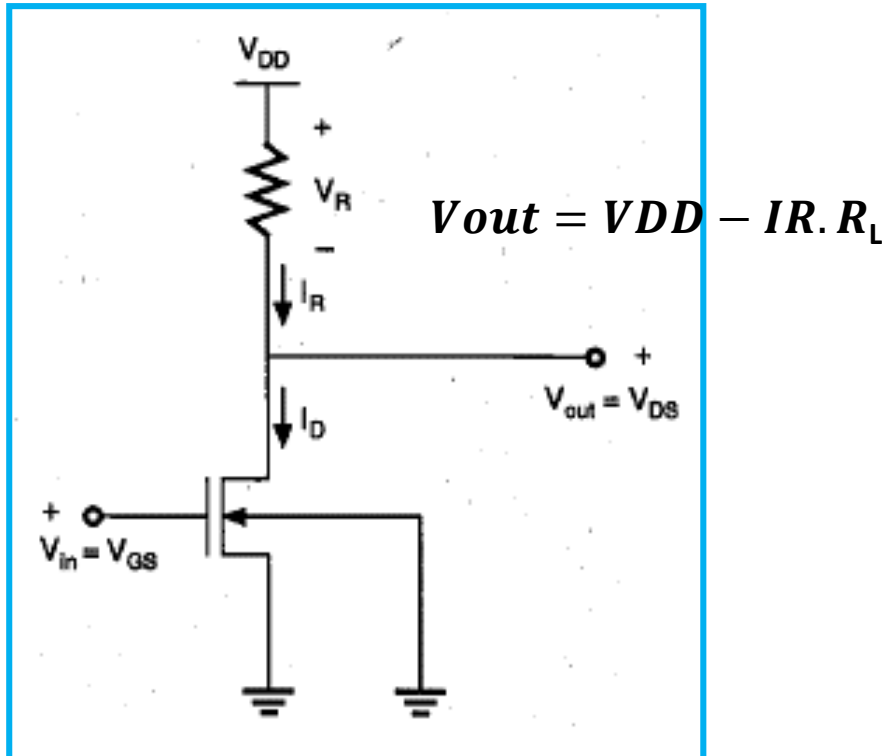
Part c - Resistive Load Inverter

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MOS Inverters (Static Characteristics)

Schematic of a Resistive Load Inverter



NMOS Device (Driver)

Device in Saturation:

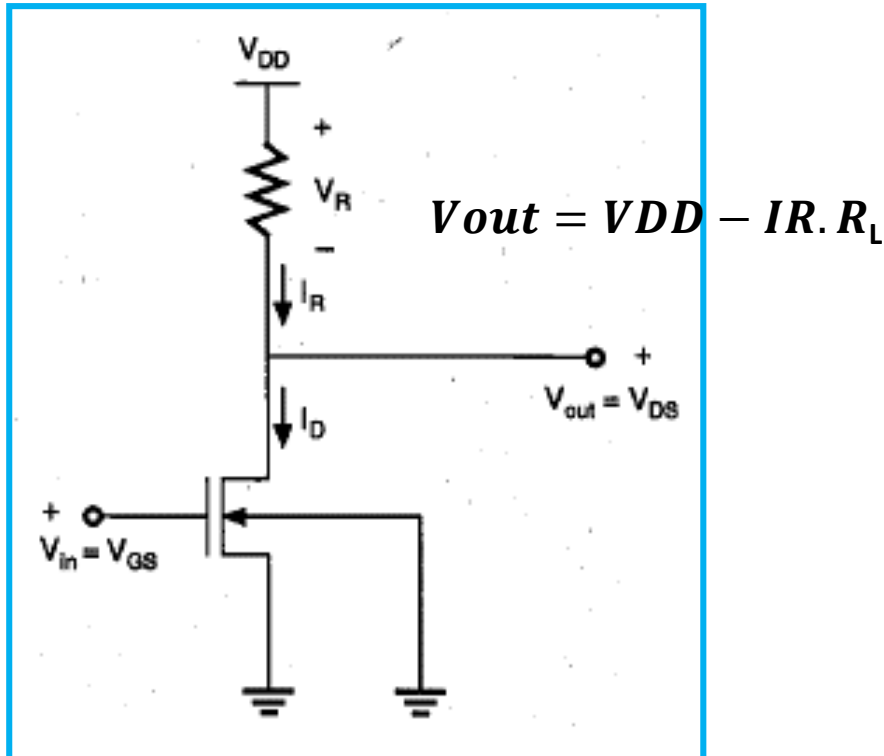
$$I_D = \frac{kn}{2} (V_{gs} - V_{tn})^2$$
$$= \frac{kn}{2} (V_{in} - V_{tn})^2$$

Device in Linear:

$$I_D = \frac{kn}{2} [2(V_{gs} - v_{tn})V_{ds} - V_{ds}^2]$$
$$= \frac{kn}{2} [2(V_{in} - v_{tn})V_{out} - V_{out}^2]$$

MOS Inverters (Static Characteristics)

Schematic of a Resistive Load Inverter



Input Voltage Range	Operating Mode
$V_{in} < V_{T0}$	cut-off
$V_{T0} \leq V_{in} < V_{out} + V_{T0}$	saturation
$V_{in} \geq V_{out} + V_{T0}$	linear

Table 5.1. Operating regions of the driver transistor in the resistive-load inverter.

MOS Inverters (Static Characteristics)

Schematic of a Resistive Load Inverter

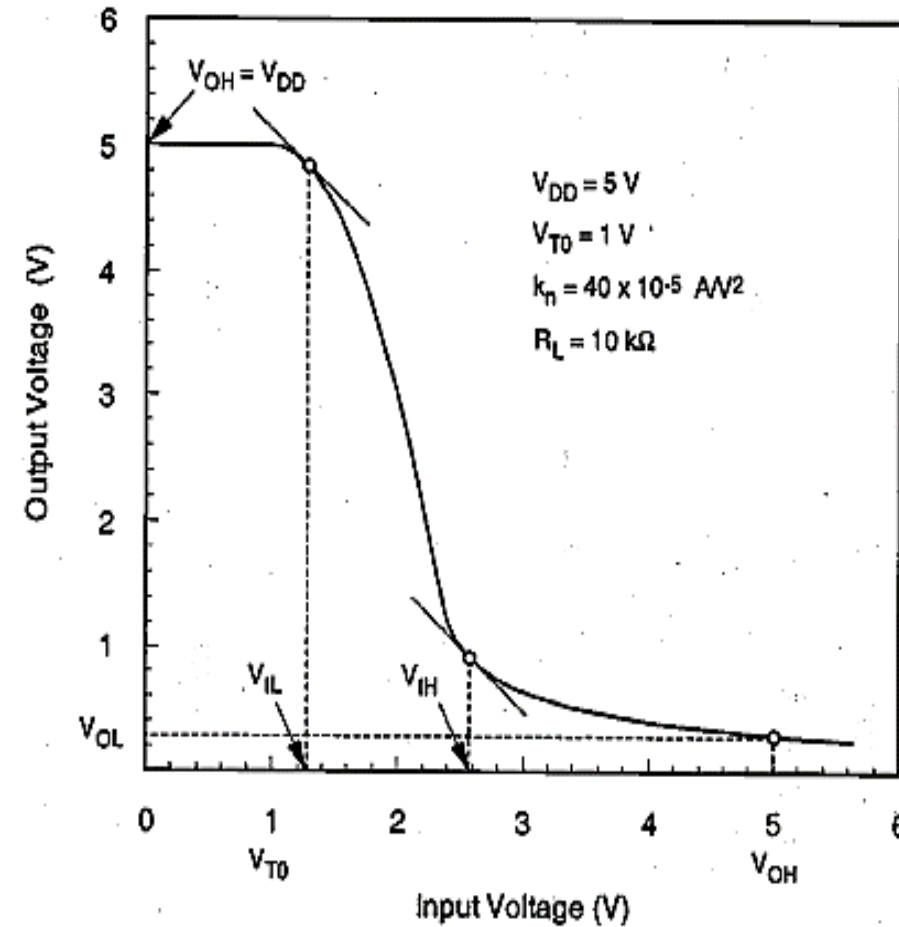
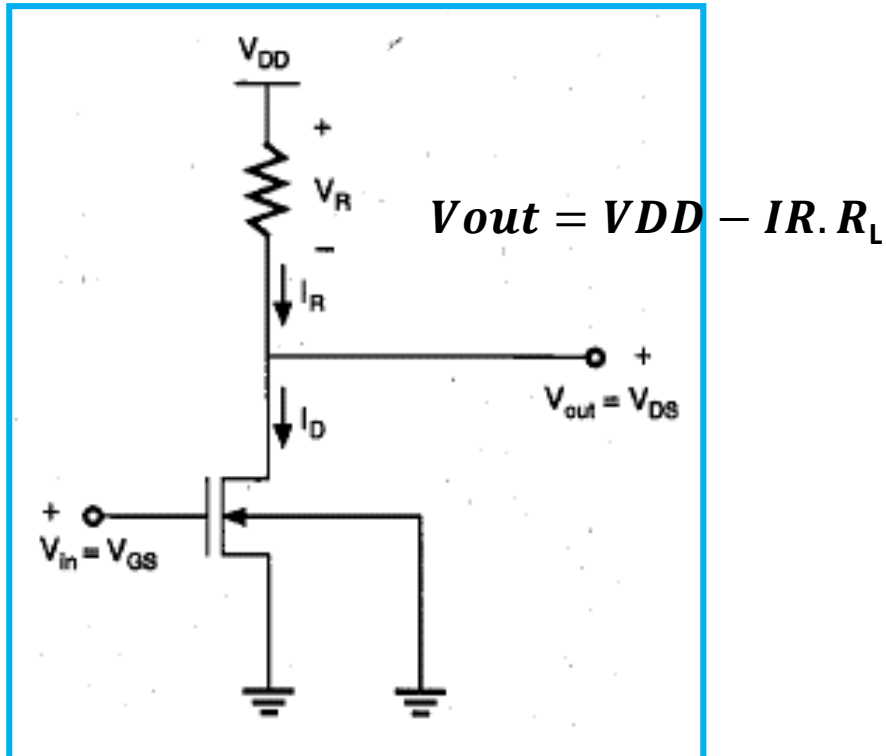


Figure 5.8. Typical VTC of a resistive-load inverter circuit. Important design parameters of the circuit are shown in the inset.

MOS Inverters (Static Characteristics)

Calculation of V_{OH}

w.k.t

$$V_{out} = V_{DD} - I_R \cdot V_R \text{ ----- (1)}$$

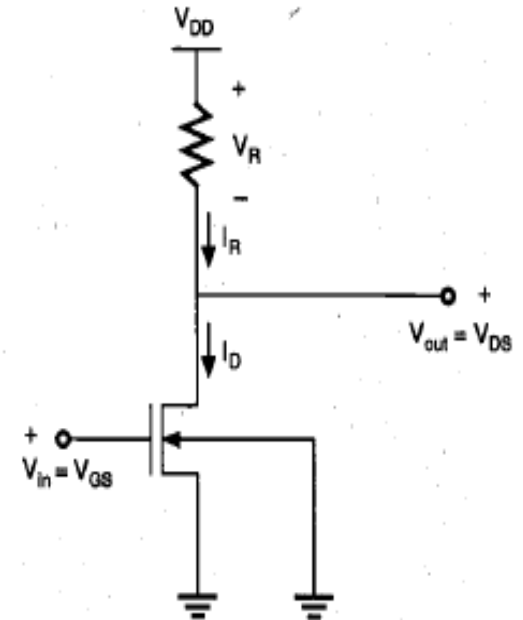
V_{OH} is the output when V_{in} is Low

($V_{in} < V_{tn}$). The *nmos* is in Cut off region. Therefore $I_D = I_R = 0$

Hence,

$$V_{out} = V_{DD} - 0 = V_{DD}$$

$$V_{out} = V_{OH} = V_{DD} \text{ ----- (2)}$$



MOS Inverters (Static Characteristics)

Calculation of V_{OL}

Consider $V_{in} = V_{OH} = V_{dd}$, the device is in **linear region**.

w.k.t $I_D = I_R$

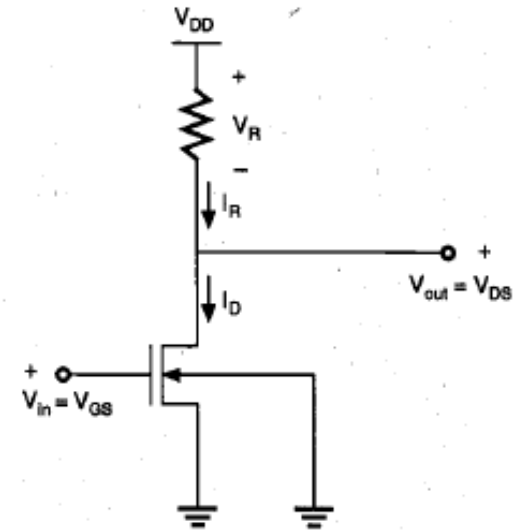
$$I_R = \frac{(V_{dd} - V_{out})}{R_L} = \frac{(V_{dd} - V_{OL})}{R_L} \text{ ----- (1)}$$

$$\begin{aligned} I_D &= \frac{k_n}{2} [2(V_{gs} - V_{T0})V_{ds} - V_{ds}^2] \\ &= \frac{k_n}{2} [2(V_{in} - V_{T0})V_{OL} - V_{OL}^2] \text{ ----- (2)} \end{aligned}$$

Therefore, by KCL

$$\frac{(V_{dd} - V_{OL})}{R_L} = \frac{k_n}{2} [2(V_{in} - V_{T0})V_{OL} - V_{OL}^2] \text{ --- (3)}$$

By rearranging



MOS Inverters (Static Characteristics)

Consider $V_{in} = V_{OH} = V_{dd}$, the device is in **linear region**.

w.k.t $I_D = I_R$

$$I_R = \frac{(V_{dd} - V_{out})}{R_L} = \frac{(V_{dd} - V_{OL})}{R_L} \text{ ----- (1)}$$

$$\begin{aligned} I_D &= \frac{k_n}{2} [2(V_{gs} - V_{T0})V_{ds} - V_{ds}^2] \\ &= \frac{k_n}{2} [2(V_{in} - V_{T0})V_{OL} - V_{OL}^2] \text{ ----- (2)} \end{aligned}$$

Therefore, by KCL

$$\frac{(V_{dd} - V_{OL})}{R_L} = \frac{k_n}{2} [2(V_{in} - V_{T0})V_{OL} - V_{OL}^2] \text{ --- (3)}$$

By rearranging

$$V_{OL}^2 - 2(V_{in} - V_{T0})V_{OL} + \frac{(V_{dd} - V_{OL})^2}{R_L \cdot k_n} = 0$$

$$V_{OL}^2 - 2(V_{in} - V_{T0})V_{OL} + \frac{2 \cdot V_{dd}}{R_L \cdot k_n} - \frac{2 \cdot V_{OL}}{R_L \cdot k_n} = 0$$

MOS Inverters (Static Characteristics)

$$V_{OL}^2 - 2 \left(V_{dd} - V_{T_0} + \frac{1}{kn \cdot RL} \right) V_{OL} + \frac{2V_{dd}}{kn \cdot RL} = 0 \quad \text{--- (4)}$$

$$V_{OL} = \frac{2 \left(V_{dd} - V_{T_0} + \frac{1}{kn \cdot RL} \right) - \sqrt{4 \left(V_{dd} - V_{T_0} + \frac{1}{kn \cdot RL} \right)^2 - 4 \frac{2V_{dd}}{kn \cdot RL}}}{2}$$

$$V_{OL} = \left(V_{dd} - V_{T_0} + \frac{1}{kn \cdot RL} \right) - \sqrt{\left(V_{dd} - V_{T_0} + \frac{1}{kn \cdot RL} \right)^2 - \frac{2V_{dd}}{kn \cdot RL}}$$

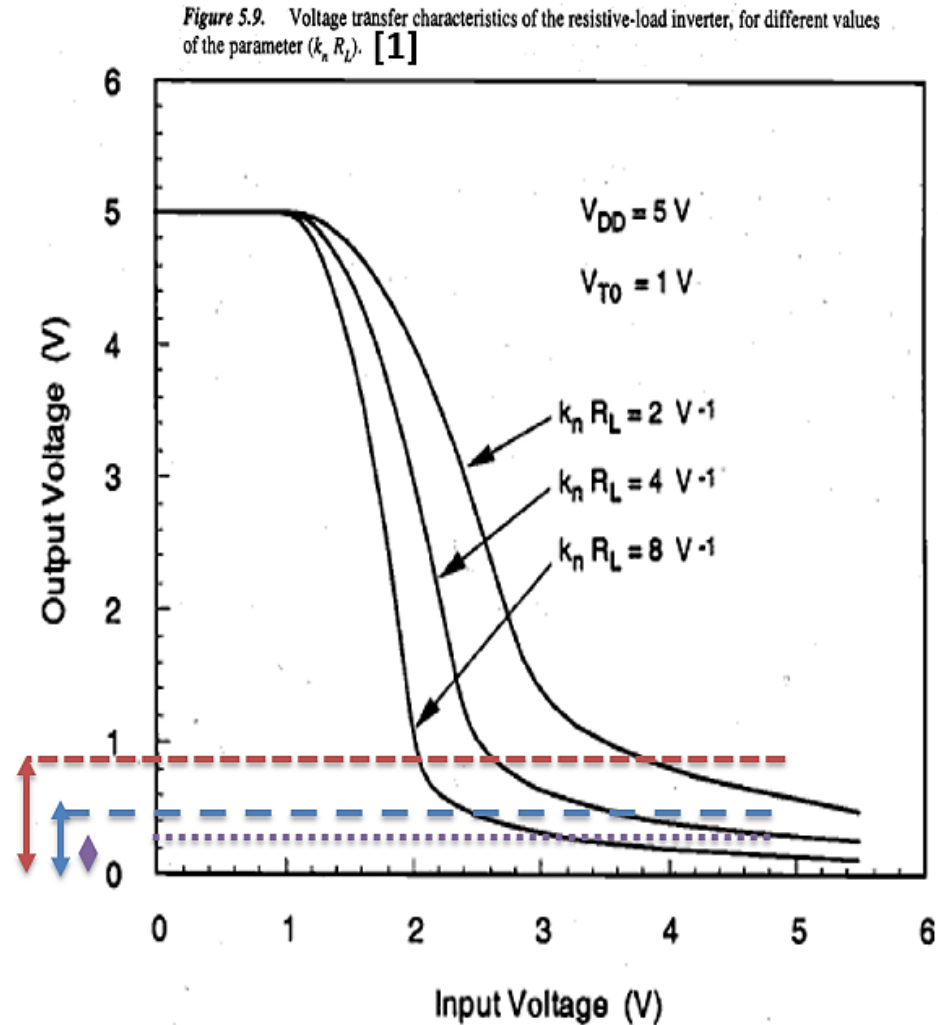
Out of two possible solutions the one which is physically correct for output low is chosen.

Observation:

Term $(kn \cdot RL)$ in V_{OL} is in the denominator. As a result is clear that if $(kn \cdot RL)$ is SMALL then V_{OL} is LARGE and As it increases V_{OL} decreases i.e., at HIGHER value of $(kn \cdot RL)$ V_{OL} is small and near to ideal value of a inverter

MOS Inverters (Static Characteristics)

Resistive Load Inverter: VTC for different values of parameter ($k_n R_L$)



MOS Inverters (Static Characteristics)

Calculation of V_{IL}

when $V_{in} = V_{IL}$,
 V_{out} is slightly smaller than V_{OH} , the device is in **saturation**

Now by KCL at output node, $I_D = I_R$

$$I_R = \frac{(V_{dd} - V_{out})}{R_L} \text{ ----- (1)}$$

$$I_D = \frac{kn}{2} (V_{gs} - V_{T0})^2 = \frac{kn}{2} (V_{in} - V_{T0})^2 \text{ ----- (2)}$$

$$\frac{(V_{dd} - V_{out})}{R_L} = \frac{kn}{2} (V_{in} - V_{T0})^2 \text{ ----- (3)}$$

By differentiating the above equation

$$\frac{dV_{out}}{dV_{in}} \left(\frac{-1}{R_L} \right) = \frac{Kn}{2} \cdot 2 \cdot (V_{in} - V_{T0})$$

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$$\frac{(V_{dd} - V_{out})}{R_L} = \frac{kn}{2} (V_{in} - V_{T0})^2 \text{ ----- (3)}$$

By differentiating the above equation

$$\frac{dV_{out}}{dV_{in}} \left(\frac{-1}{R_L} \right) = \frac{Kn}{2} \cdot 2 \cdot (V_{in} - V_{T0})$$

$$\left(\frac{-1}{R_L} \right) \frac{dV_{out}}{dV_{in}} = kn \cdot (V_{in} - V_{T0}) \text{ ---- (4)}$$

MOS Inverters (Static Characteristics)

Now, Substitute $\left(\frac{dV_{out}}{dV_{in}} = -1\right)$ & $V_{in} = V_{IL}$

$$\left(\frac{-1}{R_L}\right)(-1) = kn \cdot (V_{IL} - V_{T0})$$

$$\left(\frac{1}{knR_L}\right) = (V_{IL} - V_{T0})$$

$$V_{IL} = V_{T0} + \frac{1}{kn \cdot RL} \text{ ----- (5)}$$

Now, Let us check what is V_{out} when $V_{in} = V_{IL}$

Substitute eqn (5) in (3)

$$\frac{(V_{dd} - V_{out})}{RL} = \frac{kn}{2} \left(V_{T0} + \frac{1}{kn \cdot RL} - V_{T0} \right)^2$$

MOS Inverters (Static Characteristics)

By simplifying and separating Vout terms

$$\frac{V_{dd}}{R_L} - \frac{V_{out}}{R_L} = \frac{kn}{2} \left(\frac{1}{kn \cdot RL} \right)^2$$

$$\frac{V_{out}}{R_L} = \frac{V_{dd}}{R_L} - \frac{kn}{2} \left(\frac{1}{kn \cdot RL} \right)^2$$

$$V_{out} = V_{dd} - \left(\frac{1}{2kn \cdot RL} \right) \text{-----(6)}$$

MOS Inverters (Static Characteristics)

Calculation of V_{IH}

When $V_{in}=V_{IH}$, V_{out} is slightly larger than V_{OL} and the device is in liner region.

By KCL,

$$\frac{(V_{dd}-V_{out})}{R_L} = \frac{kn}{2} [2(V_{in} - V_{TO})V_{out} - V_{out}^2] \text{----(1)}$$

As V_{IH} is defined at slope =(-1), let us differentiate eqn (1) w.r.t V_{in}

$$-\frac{1}{R_L} \cdot \frac{dV_{out}}{dV_{in}} = \frac{kn}{2} \left[2(V_{in} - V_{TO}) \frac{dV_{out}}{dV_{in}} + 2V_{out} - 2V_{out} \cdot \frac{dV_{out}}{dV_{in}} \right] \text{-(2)}$$

Now, Substitute $\left(\frac{dV_{out}}{dV_{in}} = -1 \right)$ & $V_{in} = V_{IH}$

When $V_{in}=V_{IH}$, V_{out} is slightly larger than V_{OL} and the device is in liner region.

By KCL,

$$\frac{(V_{dd}-V_{out})}{R_L} = \frac{kn}{2} [2(V_{in} - V_{TO})V_{out} - V_{out}^2] \text{----(1)}$$

As V_{IH} is defined at slope $=(-1)$, let us differentiate eqn (1) w.r.t V_{in}

$$-\frac{1}{R_L} \cdot \frac{dV_{out}}{dV_{in}} = \frac{kn}{2} \left[2(V_{in} - V_{TO}) \frac{dV_{out}}{dV_{in}} + 2V_{out} - 2V_{out} \cdot \frac{dV_{out}}{dV_{in}} \right] \text{-(2)}$$

Now, Substitute $\left(\frac{dV_{out}}{dV_{in}} = -1 \right)$ & $V_{in} = V_{IH}$

$$-\frac{1}{R_L} \cdot (-1) = \frac{kn}{2} [2(V_{IH} - V_{TO})(-1) + 2V_{out} - 2V_{out} \cdot (-1)]$$

$$\frac{1}{R_L} = kn[(V_{IH} - V_{TO})(-1) + V_{out} + V_{out}.]$$

$$\frac{1}{kn.RL} = [-V_{IH} + V_{T0} + 2V_{out}]$$

$$V_{IH} = \left[V_{T0} + 2V_{out} - \frac{1}{kn.RL} \right] \text{ ----- (3)}$$

In eqn (3), w. k. t V_{out} is the voltage when $V_{in} = V_{IH}$
Therefore let us find V_{out} by substituting (3) in (1)

$$\frac{(V_{dd} - V_{out})}{R_L} = \frac{kn}{2} \left[2 \left(V_{T0} + 2V_{out} - \frac{1}{kn.RL} \right) V_{out} - V_{out}^2 \right]$$

$$\frac{(V_{dd} - V_{out})}{R_L} = \frac{kn}{2} \left[2 \left(2V_{out} - \frac{1}{kn.RL} \right) V_{out} - V_{out}^2 \right]$$

By rearranging

$$V_{out}^2 - 2 \left(2V_{out} - \frac{1}{kn.RL} \right) V_{out} + \frac{2.(V_{dd} - V_{out})}{kn.RL} = 0$$

MOS Inverters (Static Characteristics)

$$V_{out}^2 - 2 \left(2V_{out} - \frac{1}{kn \cdot RL} \right) V_{out} + \frac{2 \cdot V_{dd}}{kn R_L} - \frac{2V_{out}}{kn RL} = 0$$

$$V_{out}^2 - 2 \left(2V_{out} - \frac{1}{kn \cdot RL} + \frac{1}{kn RL} \right) V_{out} + \frac{2 \cdot V_{dd}}{kn R_L} = 0$$

$$V_{out}^2 - 2(2V_{out})V_{out} + \frac{2 \cdot V_{dd}}{kn R_L} = 0$$

$$-3V_{out}^2 + \frac{2 \cdot V_{dd}}{kn R_L} = 0$$

$$V_{out}^2 = \frac{2 \cdot V_{dd}}{3 \cdot kn RL}$$

$$V_{out} = \sqrt{\frac{2 \cdot V_{dd}}{3 \cdot kn RL}} \text{ -----(4)}$$

MOS Inverters (Static Characteristics)

Now substitute (4) in (3)

$$V_{IH} = \left[V_{T0} + 2 \cdot \sqrt{\frac{2.V_{dd}}{3.knRL}} - \frac{1}{kn.RL} \right]$$

$$V_{IH} = \left[V_{T0} + \sqrt{\frac{8.V_{dd}}{3.knRL}} - \frac{1}{kn.RL} \right] \text{----- (5)}$$

Observation:

Term $(kn.RL)$ in V_{IH} is in the denominator. This term affects V_{IH}

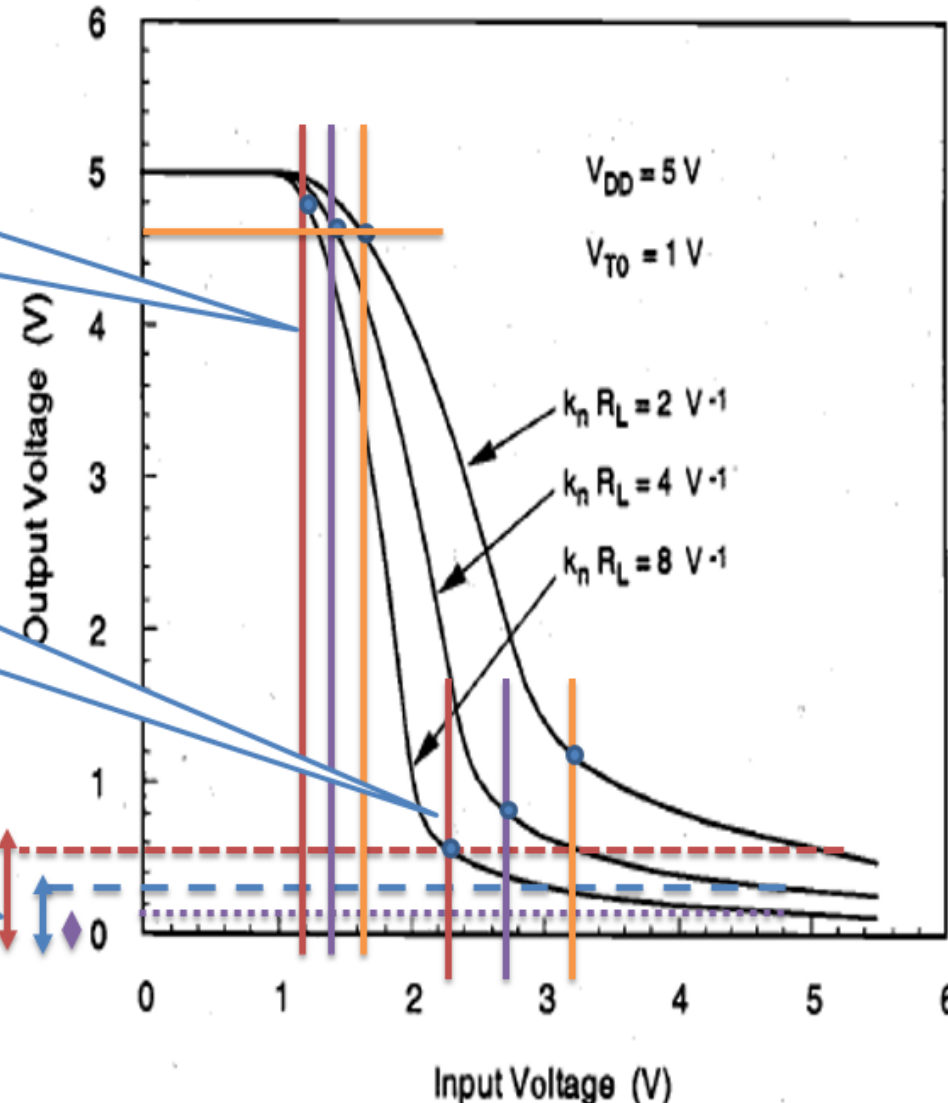
MOS Inverters (Static Characteristics)

Resistive Load Inverter: VTC for different values of parameter ($k_n R_L$)

Higher the value of $k_n R_L$
Smaller is the V_{IL} value.
i.e logic 0 – Small range

Higher the value of $k_n R_L$
Smaller is the V_{IH} value
i.e logic 1 – Larger range

Higher the value of $k_n R_L$
Smaller is the V_{OL} value
i.e For ideal value of V_{OL} of
inverter the $k_n R_L$ should be
larger



MOS Inverters (Static Characteristics)

Power Consumption and chip area

Power : $P_{DC} = V_{DD} \cdot I_{DC}$

If the duty cycle of input pulse is 50% then

$$P_{DC} = \frac{V_{DD}}{2} [I_{DC} (V_{in} = Low) + I_{DC} (V_{in} = High)]$$

In Resistive load inverter, When $V_{in} = Low (VOL)$, nmos is OFF ($I_D = 0 = I_R$) and when $V_{in} = VOH (HIGH)$, nmos is in linear region therefore both conduct nonzero current. The current drawn

$$I_D = I_R = \frac{(V_{DD} - V_{out})}{R_L} = \frac{(V_{DD} - VOL)}{R_L}$$

$$P_{DC} = \frac{V_{DD}}{2} [0 + I_{DC} (V_{in} = High)]$$

$$P_{DC} = \frac{V_{DD}}{2} \left(\frac{V_{DD} - VOL}{R_L} \right)$$

MOS Inverters (Static Characteristics)

Chip Area

Chip area occupied by Resistive load Inverter depends on (W/L) ratio of the driver transistor and value of the load resistor.

a) The area of the driver transistor can be approximately equal to gate area ($W \times L$). Assuming L is kept minimum according to technology, the gate area will be proportional to (W/L) ratio of driver transistor.

b) The resistor area, on the other hand depends on technology used to fabricate the resistor. The technologies used for resistor fabrication are

i) Diffused resistor: Isolated n-type (or p type) diffusion region with one contact on each end. **The resistance is determined by density of the diffusion region and the dimensions i.e, (L/W) ratio of resistor.**

Diffusion region sheet resistance range form **20 to 100 ohm/ square**. Consequently, very large length to width ratios would be required to achieve resistor (10-100)kohm.

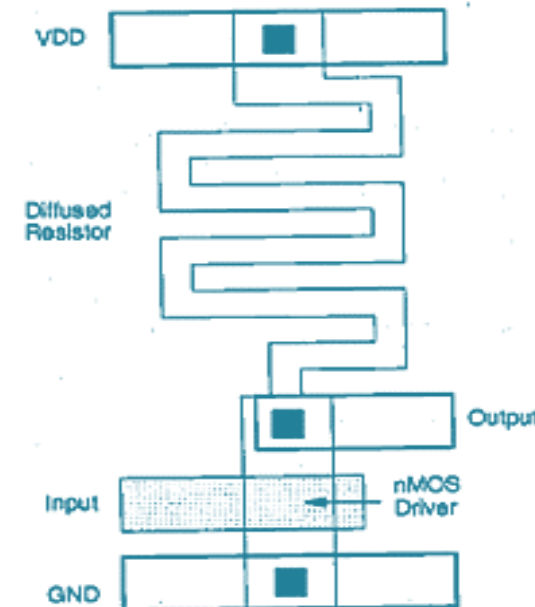


Figure 5.10. Sample layout of resistive-load inverter circuits with (a) diffused resistor and (b) undoped polysilicon resistor.

MOS Inverters (Static Characteristics)

ii) Polysilicon (un-doped) resistor: Polysilicon layers are heavily doped in order to reduce resistivity. The sheet resistivity of doped polysilicon interconnect and gate is 20-40 ohms/square.

If the section of polysilicon of masked form doping i.e., resulting un-doped polysilicon results in very high sheet resistivity of the order 10Mohm/square. Thus compact and very high valued resistor can be fabricated by using a small area.

Drawback: Resistance values of un-doped polysilicon can not be controlled very accurately, which results in large variation in VTC. Therefore resistive load inverter with un-doped poly are not commonly used for fabrication of logic gate circuits.

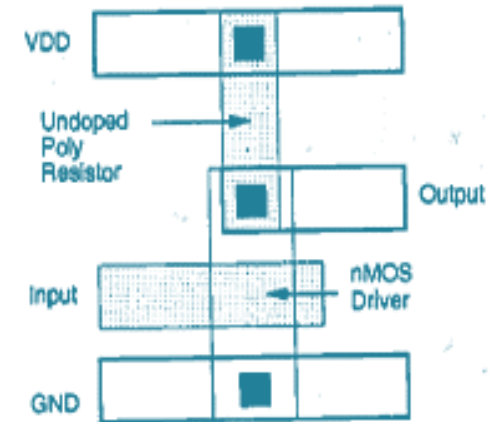


Figure 5.10. Sample layout of resistive-load inverter circuits with (a) diffused resistor and (b) undoped polysilicon resistor.

MOS Inverters (Static Characteristics)

Example 5.1

Consider the following inverter design problem: Given $V_{DD} = 5\text{ V}$, $k_n' = 30\text{ }\mu\text{A/V}^2$, and $V_{T0} = 1\text{ V}$, design a resistive-load inverter circuit with $V_{OL} = 0.2\text{ V}$. Specifically, determine the (W/L) ratio of the driver transistor and the value of the load resistor R_L that achieve the required V_{OL} .

$$\frac{(V_{DD}-V_{OL})}{R_L} = \frac{kn'}{2} \left(\frac{w}{L}\right) [2(V_{in} - V_{T0})V_{OL} - V_{OL}^2]$$

$$\frac{(V_{DD}-V_{OL})}{R_L} = \frac{kn'}{2} \left(\frac{w}{L}\right) [2(V_{OH} - V_{T0})V_{OL} - V_{OL}^2]$$

$$\frac{(5-0.2)}{R_L} = \frac{30 \times 10^{-6}}{2} \left(\frac{w}{L}\right) [2(5 - 1)0.2 - 0.2^2]$$

$$\left(\frac{w}{L}\right) R_L = 2.05 \times 10^5$$

MOS Inverters (Static Characteristics)

$\left(\frac{W}{L}\right)$ - Ratio	Load Resistor R_L [k Ω]	DC Power Consumption $P_{DC,average}$ [μ W]
1	205.0	58.5
2	102.5	117.1
3	68.4	175.4
4	51.3	233.9
5	41.0	292.7
6	34.2	350.8

Power consumption increases as the value of the load resistor R_L is decreases and (W/L) increases.

If Power consumption is the concern, (W/L) is kept LOW and a large load resistor (R_L) need to be considered but large resistor will require more area of silicon for fabrication.

Example 5.2

Consider a resistive-load inverter circuit with $V_{DD} = 5\text{ V}$, $k_n' = 20\text{ }\mu\text{A/V}^2$, $V_{TD} = 0.8\text{ V}$, $R_L = 200\text{ k}\Omega$, and $W/L = 2$. Calculate the critical voltages (V_{OL} , V_{OH} , V_{IL} , V_{IH}) on the VTC and find the noise margins of the circuit.

$$V_{OL} = 0.147\text{ V}, V_{OH} = 5\text{ V}, V_{IL} = 0.925\text{ V}, V_{IH} = 1.97\text{ V}, \\ NM_L = 0.78\text{ V}, NM_H = 3.03$$

Observations: NML is quite Low and which may lead to misinterpretation of input signal level.

For better noise immunity, the noise margin for 'LOW' signal should be at least 25% of VDD i.e 1.25V for VDD=5V

Exercise Problem 5.1 – Similar to Example 5.1

5.1 Design a resistive-load inverter with $R = 1 \text{ k}\Omega$, such that $V_{OL} = 0.6 \text{ V}$.

The enhancement-type nMOS driver transistor has the following parameters:

$$V_{DD} = 5.0 \text{ V}$$

$$V_{T0} = 1.0 \text{ V}$$

$$\gamma = 0.2 \text{ V}^{1/2}$$

$$\lambda = 0$$

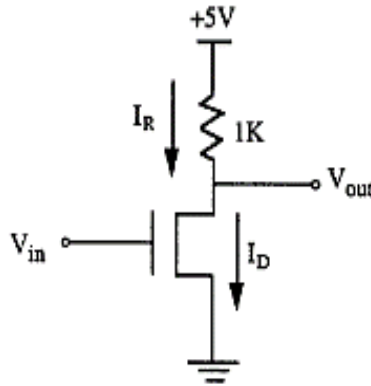
$$\mu_n C_{ox} = 22.0 \text{ }\mu\text{A/V}^2$$

- (a) Determine the required aspect ratio, W/L .
- (b) Determine V_{IL} and V_{IH} .
- (c) Determine noise margins NM_L and NM_H .

MOS Inverters (Static Characteristics)

SOLUTION:

(a)



When $V_{out} = V_{OL}$, $V_{in} = V_{DD} = 5.0V$, the driver transistor operates in linear region. Using Eq.(5.12)

$$I_R = \frac{V_{DD} - V_{OL}}{1K} = \frac{k'}{2} \frac{W}{L} [2(V_{OH} - V_{T0})V_{OL} - V_{OL}^2]$$

$$\frac{5 - 0.6}{1000} = \frac{22 \times 10^{-6}}{2} \frac{W}{L} [2(5 - 1)0.6 - 0.6^2]$$

Solve for W/L ,

$$\frac{W}{L} = 90.1$$

$$V_{IL} = V_{T0} + \frac{1}{R_L \cdot k' \frac{W}{L}} = 1.0 + \frac{1}{1000 \cdot 22 \times 10^{-6} \cdot 90.1} = 1.5[V]$$

$$V_{IH} = V_{T0} + \sqrt{\frac{8}{3} \cdot \frac{V_{DD}}{R_L \cdot k' \frac{W}{L}} - \frac{1}{R_L \cdot k' \frac{W}{L}}} \\ = 1.0 + \sqrt{\frac{8}{3} \frac{5}{1000 \cdot 22 \times 10^{-6} \cdot 90.1} - \frac{1}{1000 \cdot 22 \times 10^{-6} \cdot 90.1}} = 3.09[V]$$

$$NM_L = V_{IL} - V_{OL} = 1.5 - 0.6 = 0.9[V]$$

$$NM_H = V_{OH} - V_{IH} = 5.0 - 3.09 = 1.91[V]$$



THANK YOU

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